



GRADIENT PROJECTION METHODS FOR CONTINUAL LEARNING IN HYPERNETWORKS

OVERCOMING CHUNK-INDUCED GRADIENT
PATHOLOGIES VIA PROJECTION
NORMALIZATION, AND THE INTRODUCTION OF
IFOPNG

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Abstract

This thesis resolves the fundamental incompatibility of applying gradient projection-based continual learning (CL) methods to hypernetwork architectures, establishing a novel framework for stable subspace optimization. Traditional projection methods collapse due to a structural-numerical pathology where the architectural necessity of weight-chunking—required to avoid an output parameter explosion—forces the model to be produced in hundreds or thousands of sequential chunks before a single projection step can execute. This repetitive chunk-generation loop accumulates an exceptionally large gradient volume within the memory matrix, blowing up the condition number of the projection correlation matrix to infinity and causing catastrophic inversion failures that the standard regularization parameter λ cannot rescue.

The primary research objective is to eliminate these numerical singularities and systematically evaluate whether combining hard geometric projections on top of task-conditioned hypernetworks and their output-scoped functional regularizers yields a constructive synergy. To achieve this, we introduce *projection-scoped normalization*: gradient columns stored in the memory matrix G are rescaled to unit ℓ_2 norm, and the diagonal empirical Fisher vector is rescaled by its absolute maximum, together stabilizing the metric tensor within the local projection algebra. We also introduce iFOPNG, an elastic variant of Fisher-Orthogonal Projected Natural Gradient Descent (Garg, Kolhe, Peng, & Gopalam, n.d.) that replaces the current-task Fisher F_{new} with the combined curvature $F_c = F_{\text{new}} + F_{\text{old}}$, inspired by the parameter-inertia concept of Elastic Weight Consolidation. Methods are benchmarked on sequential Split-MNIST and Split-CIFAR10 streams. The empirical findings reveal that projection normalization completely resolves matrix inversion collapse, and that in the hypernetwork setting all normalized projection methods outperform plain

SGD, with iFOPNG matching the best projection baseline. However, a standalone hypernetwork optimized with Adam and von Oswald’s functional regularizer consistently surpasses all projection variants, indicating that momentum—which projection methods forgo by design—is the decisive factor in compressed generative weight spaces.

1 DATA SOURCE, ETHICS, CODE, AND TECHNOLOGY STATEMENT

Data Source. Split-MNIST (owned by LeCun et al.) and Split-CIFAR10 (owned by Krizhevsky et al.) are public, anonymized datasets licensed for research. No human or animal data was collected. Data was acquired via the torchvision.datasets API.

FIGURES. All figures and plots were created entirely by the author using matplotlib and Weights & Biases.

CODE. The baseline FOPNG framework was adapted from Garg et al. (n.d.); all other execution infrastructure was implemented by the author (<https://github.com/kelares/HyperFisher>). Libraries used: Python (v3.10), PyTorch (v2.x), Torchvision, Weights & Biases, NumPy, Matplotlib, and Tqdm.

TECHNOLOGY. The manuscript was typeset in L^AT_EX using BibTeX. During the preparation of this work the author(s) used Gemini in order to optimize academic phrasing, outline structure, paraphrase transitions, and check grammar. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the thesis. AI Citation: Google. (2026). *Gemini 3 Flash* [Large language model]. <https://gemini.google.com>.

2 INTRODUCTION

The capacity for continuous, sequential adaptation without the eradication of historical knowledge remains an architectural cornerstone of biological intelligence, yet it presents a fundamental barrier for artificial neural networks. In deep learning paradigms, sequential training across non-stationary data distributions triggers a systemic pathology known as *catastrophic forgetting* (?). When a network optimizes its parameters for an incoming task, gradient updates blindly overwrite the weight configurations essential for retaining prior task spaces, shifting the internal representation manifolds away from historical objective minima. Modern continual learning (CL) literature is conventionally structured into three

core algorithmic families: replay-based strategies that leverage episodic data buffers to rehearse historical distributions, regularization-based frameworks that penalize parameter deviations along important directions, and architectural or parameter-isolation methods (De Lange et al., n.d.). Within the meta-learning domain, Hypernetworks (?) have emerged as an elegant paradigm. Hypernetworks circumvent parameter-space conflicts by utilizing a centralized, lightweight meta-model to generate the target network’s complete parameter weights conditioned on task embeddings, as formalized by Oswald, Henning, Grewe, and Sacramento (n.d.). Concurrently, within the field of geometric optimization, gradient projection methods—exemplified by the recently proposed Fisher-Orthogonal Projected Natural Gradient Descent (FOPNG) (Garg et al., n.d.)—explicitly constrain subsequent updates to the null-space of historical parameter spaces.

Despite the mathematical elegance of both hypernetworks and gradient projection methods, their structural intersection remains almost entirely unexamined. Crucially, this investigation uncovers a critical hidden optimization pathology born at this intersection. To prevent a catastrophic parameter explosion within the meta-model, a hypernetwork must use sequential weight-chunking; generating the entire target weight array simultaneously would require an absurdly large output layer, defeating the purpose of a compact meta-learner. Depending on the dimensional footprint of the target architecture, this constraint forces the model to be produced in hundreds or thousands of individual chunks. Running this sub-block spawning mechanism completely prior to a single execution step accumulates gradients additively, resulting in parameter gradients of exceptionally high magnitude. This explosion completely erases the stabilizing effect of λ during matrix inversion, blowing up the condition number and inducing catastrophic matrix inversion failures.

The primary research objective of this thesis is to systematically resolve this linear-algebraic collapse, map the initialization mechanics of meta-projection pipelines, and evaluate whether the integration of hard geometric constraints on top of soft functional regularizers yields an optimized sequential learning system. To achieve this, we introduce an algorithmic framework that unifies three core elements: (i) a local projection-scoped normalization mechanism that scales gradient vectors and empirical Fisher Information Matrix (FIM) components strictly within the projection scope; (ii) a pure decoupled initialization protocol via AdamW to prevent baseline gradient contamination; and (iii) Elastic FOPNG (iFOPNG), a custom projection variant inspired by the concept of parameter inertia from EWC. Our framework is benchmarked across sequential multi-head streams using Split-MNIST and Split-CIFAR10 distributions.

The scientific relevance of this thesis lies in identifying and resolving numerical optimization anomalies at the intersection of gradient projection methods and chunked hypernetworks. First, we demonstrate that additive gradient accumulation across thousands of sequential sub-chunks creates extreme gradient magnitudes that numerically erase the stabilizing effect of λ during matrix inversion (Garg et al., n.d.). We resolve this by developing projection-scoped normalization: column-wise ℓ_2 normalization of the gradient matrix G and absolute-maximum scaling of the diagonal empirical Fisher vector. Second, this work establishes a rigorous baseline for initializing generative geometric subspaces, demonstrating that AdamW eliminates the structural gradient contamination induced by standard Adam with L_2 regularization. Third, we introduce iFOPNG as a novel algorithmic variant that replaces the rigid orthogonal boundary of FOPNG with an elastic curvature boundary formed by the combined Fisher $F_c = F_{\text{new}} + F_{\text{old}}$.

The societal relevance of optimizing these architectures extends to the real-world deployment of edge computing, autonomous robotics, and localized medical diagnostics. Enabling compact, VRAM-efficient models to permanently acquire sequential streams of knowledge on-device—without access to centralized historical data—provides a viable pathway toward secure, sustainable artificial intelligence.

To systematically navigate these boundaries, this thesis addresses the following primary research question:

To what extent can gradient projection methods be successfully integrated into chunked hypernetwork architectures for continual learning, and how do their structural pathologies, initialization protocols, and elastic variants impact optimization efficiency?

This is decomposed into four sub-research questions (Sub-RQs):

- Sub-RQ1:** *Does the joint integration of gradient projection frameworks and soft output-scoped functional regularization yield a constructive synergy, or does a standalone regularized hypernetwork offer a more efficient optimization frontier?*
- Sub-RQ2:** *How does the architectural necessity of spawning thousands of sequential weight-chunks impact gradient magnitudes, and to what extent can local projection-scoped normalization prevent the resulting explosion from collapsing the matrix inversion?*
- Sub-RQ3:** *To what degree does evaluating the natural gradient projection under a joint Riemannian metric ($F_c = F_{\text{new}} + F_{\text{old}}$), rather than the task-specific metric (F_{new}) used in standard FOPNG, influence task retention?*
- Sub-RQ4** (Fisher Accumulation Operator). Given an elastic Fisher preconditioner, how should the historical component F_{old} be accumulated across

tasks? Specifically, does the non-decaying maximum operator $F_{\text{old}} \leftarrow \max(F_{\text{old}}, F_{\text{new}})$ preserve early-task knowledge more effectively than a decaying exponential moving average, and does its monotonic growth impose a measurable plasticity cost over long task sequences?

The empirical findings reveal a definitive structural boundary: projection normalization completely resolves matrix inversion collapse; AdamW initialization establishes a cleaner base manifold than Adam; all projection variants surpass plain SGD in the hypernetwork setting, confirming a partial constructive synergy. However, a standalone hypernetwork optimized via Adam with von Oswald’s functional regularizer consistently matches or outperforms all projection variants. This indicates that the momentum mechanism—which projection optimizers forgo by design because the projection step discards the optimizer state—is the decisive factor in compressed generative weight spaces.

3 RELATED WORK

This section builds the theoretical and empirical context for the thesis, tracing a path from foundational continual learning challenges through regularization and projection methods, to hypernetwork architectures and the structural gap at their intersection.

3.1 Continual Learning: Taxonomy and the Stability-Plasticity Dilemma

Modern machine learning systems are predominantly trained on static, i.i.d. datasets — a condition that rarely holds in deployment, where agents must adapt to non-stationary data streams while retaining prior knowledge. The fundamental tension between these two objectives is formalized as the *stability-plasticity dilemma* (Grossberg, n.d.): an agent must remain plastic enough to absorb novel distributions yet stable enough to resist the erasure of established representations. When standard architectures trained by empirical risk minimization are exposed to sequential data, they suffer *catastrophic forgetting* (?) — abrupt and near-total loss of previously learned capabilities upon exposure to new task data.

The continual learning literature addresses this across three canonical scenarios (De Lange et al., n.d.): *task-incremental learning*, where the task identity is available at both training and test time; *class-incremental learning*, where task identity is withheld at inference; and *domain-incremental learning*, where the output space is shared but the input distribution shifts continuously. This thesis focuses on the task-incremental setting, the scenario in which projection-based methods and hypernetwork architectures

have been most thoroughly evaluated and in which structural comparisons between them are most tractable.

3.2 Evaluation Protocols: Multi-Head and Single-Head Designs

Within the task-incremental setting, two evaluation protocols govern how the network’s output layer is organized. In the *multi-head* protocol, each task is assigned a dedicated output head; at test time the correct head is selected using the known task identity. This eliminates output-level interference between tasks but requires that task identity be available at inference time, a condition that may be unrealistic in deployment. In the *single-head* protocol, all tasks share one output layer, requiring the backbone alone to resolve inter-task interference — a substantially harder problem.

Farquhar and Gal (n.d.) demonstrated that multi-head evaluations systematically inflate apparent method performance by removing a major source of forgetting, and argued that single-head evaluation is the more rigorous protocol for assessing continual learning efficacy. This distinction has direct implications for how gradient projection methods and hypernetworks are evaluated: projection methods were predominantly developed and benchmarked in multi-head settings, while hypernetworks implement an implicit form of task separation through their generative structure.

3.3 Regularization-Based Approaches

The dominant paradigm for addressing catastrophic forgetting is *parameter regularization*, of which Elastic Weight Consolidation (EWC) (Kirkpatrick et al., n.d.) is the canonical representative. EWC estimates the structural importance of each weight using the diagonal of the empirical Fisher Information Matrix $\hat{F}$, and applies a soft quadratic penalty that resists changes to parameters important to prior tasks:

$$\mathcal{L}_{\text{EWC}}(\theta) = \mathcal{L}_t(\theta) + \frac{\lambda}{2} \sum_i \hat{F}_i (\theta_i - \theta_{i,t-1}^*)^2. \quad (1)$$

EWC is computationally efficient and broadly applicable, and serves as the standard reference point against which new continual learning methods are measured. Its principal limitation is that soft penalties permit bounded parameter drift: cumulative updates across a long task sequence gradually erode the protection of earlier task spaces, a failure mode that becomes increasingly pronounced as T grows (Farquhar & Gal, n.d.).

A related limitation concerns baseline comparisons. The superiority of complex regularization schemes over vanilla optimizers is frequently overstated when those baselines are under-tuned. When SGD and Adam

baselines are tuned to their optimal hyperparameter regime, they often present a competitive threshold that challenges or matches more sophisticated approaches — an empirical reality that our baseline evaluations directly address.

3.4 Gradient Projection Methods

To enforce strict, non-negotiable protection for prior knowledge rather than relying on penalties that can be overcome by sufficiently large gradients, a second paradigm uses explicit *gradient projection*. Orthogonal Gradient Descent (OGD) (Farajtabar, Azizan, Mott, & Li, n.d.) maintains a memory matrix $G = [g_1 \mid \cdots \mid g_K]$ of historical task gradient directions and projects incoming gradients onto the orthogonal complement of $\text{Col}(G)$ before each update:

$$\tilde{g} = g - G(G^\top G)^{-1}G^\top g. \quad (2)$$

This provides a hard guarantee: the update cannot increase the loss on previously seen tasks. Generalization bounds established within the Neural Tangent Kernel regime confirm that OGD provides exact protection in linear and mildly nonlinear settings (Bennani, Doan, & Sugiyama, n.d.).

The mathematical appeal of projection methods over EWC lies precisely in this hardness: where EWC applies a soft penalty that can be overwhelmed, projection methods impose a geometric constraint that is either satisfied or not. However, standard Euclidean projection treats the parameter space as flat and assigns equal importance to all directions. Fisher-Orthogonal Projected Natural Gradient Descent (FOPNG) (Garg et al., n.d.) addresses this by performing the projection in the Riemannian geometry defined by the Fisher metric, so that the orthogonality constraint respects the curvature of the loss surface. The related Fisher Natural Gradient (FNG) (Garg et al., n.d.) applies the natural gradient without projection, and Orthogonal Natural Gradient (ONG) (?) combines natural gradient preconditioning with Euclidean projection — sitting structurally between OGD and FOPNG.

3.5 Hypernetworks for Continual Learning

An orthogonal architectural response to catastrophic forgetting replaces the multi-head output design entirely with a *hypernetwork* (?): a meta-model $\mathcal{H}(\cdot; \Theta_h)$ trained to synthesize the complete weight vector $\theta_t = \mathcal{H}(e_t; \Theta_h)$ of a target network, conditioned on a task embedding e_t . Each task receives a unique weight configuration generated on demand, rather than a dedicated output head, achieving implicit task separation through the generative

bottleneck and impressive parameter compression ratios relative to storing separate networks per task.

Generating the full θ_t in a single forward pass is computationally intractable for large target networks, so Oswald et al. (n.d.) introduced *weight chunking*: the generator iterates over a secondary chunk embedding c_k , producing C_s target parameters per pass and assembling the full weight vector by concatenation. To prevent the hypernetwork from forgetting how to generate prior task configurations, a soft *functional regularizer* is applied in the output space of the generator:

$$\mathcal{L}_{\text{reg}} = \beta \sum_{s=1}^{t-1} \|\mathcal{H}(\Theta_h, e_s) - \theta_s^*\|_2^2, \quad (3)$$

where θ_s^* denotes the weight snapshot saved after task s . Crucially, Oswald et al. (n.d.) demonstrated that applying parameter-space regularization — specifically EWC directly on the hypernetwork parameters Θ_h — *underperforms* the functional regularizer and actively degrades retention. The reason is structural: the Fisher measured over Θ_h is a poor proxy for task importance in the generated target network, since the same hypernetwork parameters influence all tasks through the shared generative function. Functional regularization, by penalizing changes in the generated weight space directly, provides the correct inductive bias for a meta-generative architecture.

Recent extensions have refined the hypernetwork approach: Partial Hypernetworks (Hemati, Lomonaco, Bacciu, & Borth, n.d.) reduce the computational overhead of full weight generation, and Prototype Augmented Hypernetworks (Fuente et al., n.d.) augment task embeddings with prototype representations to further stabilize sequential learning.

3.6 Identified Gaps and Motivating Pathologies

Oswald’s finding raises a critical and unanswered question: if EWC-style *soft parameter penalties* are insufficient for hypernetwork continual learning, do the *hard geometric constraints* of gradient projection methods — which operate through a fundamentally different mechanism — fare differently? Projection methods do not penalize the hypernetwork’s parameter values; they constrain the *direction* of the generator’s updates. Whether this directional constraint is compatible with the hypernetwork’s generative structure, or introduces its own failure modes, has not been investigated. This intersection is the primary research contribution of this thesis. We identify five specific structural pathologies that arise at this intersection and that the experiments are designed to address.

1. **FUNCTIONAL-PARAMETRIC INTERFACE CONFLICT (SUB-RQ1).** Layering hard parameter-space projections onto a hypernetwork that already employs soft output-space regularization creates a potential structural conflict: the projection constraint strips momentum from the generator’s updates, while the functional regularizer simultaneously pulls them toward prior weight configurations. Whether this joint constraint regime yields a constructive synergy or over-constrains the generator’s plasticity remains an open empirical question.

2. **CHUNK-INDUCED GRADIENT ACCUMULATION PATHOLOGY (SUB-RQ2).** The architectural necessity of spawning hundreds or thousands of sequential weight-chunks before a single projection step executes causes gradient vectors and Fisher estimates to accumulate to extreme magnitudes, inflating the condition number of the projection kernel $G^\top \hat{F} G$ and causing matrix inversion failures that standard regularization parameters cannot mitigate. This pathology is unique to the hypernetwork setting and has no analogue in standalone network experiments.

3. **FIRST-TASK INITIALIZATION FOOTPRINT (SUB-RQ3).** The gradient memory G and Fisher estimate $\hat{F}$ that seed the projection subspace are collected entirely from the first task. Standard Adam couples L_2 weight decay into the gradient estimate, contaminating both G and $\hat{F}$ with a weight-magnitude component unrelated to task-loss curvature. AdamW’s decoupled weight decay preserves gradient geometric purity; whether this produces measurably superior projection quality is the subject of Sub-RQ3 (Hu, Yu, Cheng, Liu, & Song, n.d.).

4. **PROJECTION RIGIDITY AND ELASTIC PARAMETER INERTIA (SUB-RQ4).** Rigid projection baselines enforce an absolute boundary that completely locks historical gradient directions, stripping the optimizer of degrees of freedom that may be needed for efficient new-task learning. We address this rigidity through iFOPNG, an elastic variant of FOPNG introduced in this thesis, which replaces the hard projection boundary with a combined Fisher preconditioner $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$. Rather than imposing a hard constraint, F_c damps movement in directions important to *any* prior task via the metric geometry, providing an EWC-style elastic inertia boundary without a penalty term.

5. **FISHER ACCUMULATION OPERATOR AND LONG-HORIZON PLASTICITY (SUB-RQ5).** The historical Fisher $\hat{F}_{\text{old}}$ underlying the iFOPNG elastic boundary is maintained via a non-decaying maximum operator: $\hat{F}_{\text{old}} \leftarrow \max(\hat{F}_{\text{old}}, \hat{F}_{\text{new}})$. This operator guarantees that any parameter ever

deemed important retains that importance permanently, providing strong protection for early-task knowledge. Whether this monotonic accumulation imposes a measurable plasticity cost over long task sequences — relative to a decaying exponential moving average — and whether the two operators trade off forgetting against new-task learning differently across sequence lengths, is the subject of Sub-RQ5.

4 RESULTS

This section reports the empirical outcomes of the four sub-research questions in the order stated in the introduction. For each, we describe the experimental scope, present the key quantitative findings, and highlight the most important patterns; detailed interpretation is reserved for the Discussion. *Average accuracy* (A_T) is the primary evaluation metric, and *backward transfer* (BWT) is reported as a secondary measure of forgetting. Standalone target-network results that establish the reference behaviour of projection methods in their original, non-hypernetwork setting are provided in Appendix A (Figures ??–??).

4.1 Sub-RQ1: Constructive Synergy or Architectural Conflict?

This subsection examines whether combining gradient projection with the hypernetwork’s functional regularizer yields better continual learning than either component alone. Two hypernetwork regimes are evaluated: a standard-capacity Split-CIFAR10 network and an extremely compressed Split-MNIST network (suffocated regime, $d_h = 8$).

STANDARD HYPERNETWORK — SPLIT-CIFAR10 (EXPERIMENT 8). Figure 6 presents average accuracy and BWT for all eight methods on Split-CIFAR10 with a standard-capacity hypernetwork and full normalization enabled (see Section 5.2). Plain SGD at 84.1% establishes the unconstrained lower reference point, and Adam at 90.1% defines the upper frontier a standalone hypernetwork can reach without projection. Every projection method surpasses SGD: iFOPNG leads at 85.6%, followed by EWC (81.5%), FNG (80.9%), OGD (78.3%), FOPNG (77.5%), and ONG (67.2%), confirming a partial constructive synergy with the hypernetwork’s regularizer. However, none match Adam, which leads by 4.5 pp over the best projection method.

The BWT values reveal an instructive asymmetry: Adam has effectively zero backward transfer (BWT ≈ 0.000), while all projection methods exhibit small but nonzero forgetting (iFOPNG: -0.013 ; FOPNG: -0.015). The per-task accuracy matrix for Experiment 8 (Appendix Figure ??) shows that iFOPNG actually retains task-1 accuracy at 0.78 across the full sequence,

compared to Adam’s 0.65; the accuracy shortfall of projection methods therefore originates from weaker forward transfer to later tasks rather than greater forgetting of earlier ones.

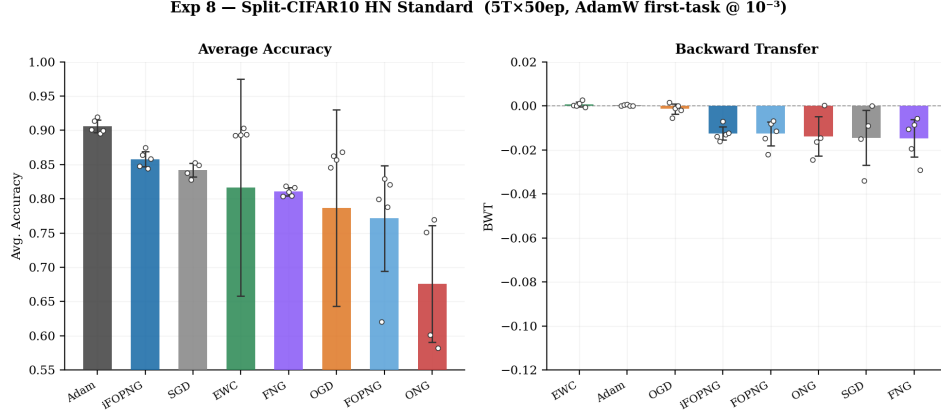


Figure 1: **Experiment 8 — Split-CIFAR10 hypernetwork (5 tasks, 50 epochs per task, AdamW first-task initialisation at 10^{-3})**. Left: average accuracy (bars show mean over seeds; circles show individual seeds). Right: backward transfer. Methods are sorted by average accuracy on the left panel. Adam defines the projection-free upper frontier; SGD defines the unconstrained lower reference. All projection methods trained with full normalization (Section 5.2) exceed SGD but fall short of Adam.

SUFFOCATED REGIME — SPLIT-MNIST, $d_h = 8$ (EXPERIMENT 7). Under extreme compression (Appendix Figure ??), the projection–Adam gap widens dramatically. Adam reaches 93.4%, while iFOPNG falls to 69.9% and FOPNG and FNG drop to 59.9% and 54.8% — gaps of 20–40 pp. Seed-level variance increases substantially for all projection methods, reflecting the highly ill-conditioned optimization landscape induced by the $d_h = 8$ bottleneck. The d_h sweep (Appendix Figure ??) confirms the effect is monotonic: the projection–Adam gap widens as d_h decreases from 16 to 8 to 4.

EXTENDED TASK COUNT — SPLIT-CIFAR100 (EXPERIMENT 11). At ten tasks on Split-CIFAR100 (Appendix Figure ??), EWC unexpectedly leads at 45.5%, with Adam second at 42.0% and iFOPNG third at 37.8% (random baseline: 10.0%). This partial rank reversal, in which soft regularization overtakes both Adam and gradient projection, differs from the five-task CIFAR10 pattern and is addressed in the Discussion.

4.2 Sub-RQ2: Normalization Resolves the Gradient Pathology

This subsection reports the three-condition normalization ablation (Experiments 8, 9, and 10). Figure 7 traces the $\log_{10}$ condition number of the projection kernel $G^\top \hat{F} G$ over training steps; Figure 8 shows the resulting average accuracy and BWT under each condition.

WITHOUT NORMALIZATION (EXPERIMENT 9). The right panel of Figure 7 shows the condition number reaching $10^{10.8}$ at the $T_2 \rightarrow T_3$ task transition for the most-affected seed, stabilizing near $10^{8.8}$ – $10^{9.0}$ for two others. At this scale, numerical inversion of the projection kernel is unreliable, effectively zeroing the projected gradient so that the method operates as an unconstrained optimizer with unstable Fisher scaling. The result is 72.2% average accuracy and $\text{BWT} = -0.070$ — worse than plain SGD (84.1%) despite identical hyperparameters. The pathology therefore not only eliminates the benefit of projection but actively degrades performance below the projection-free baseline.

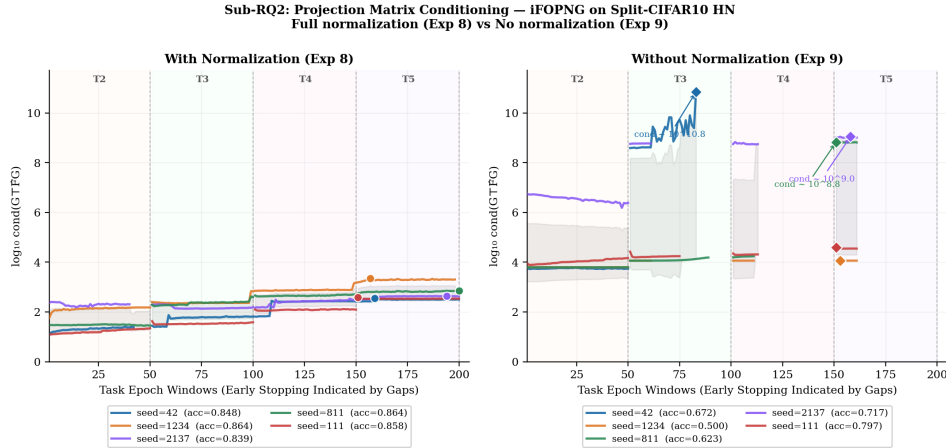


Figure 2: **Projection kernel condition number over training — iFOPNG on Split-CIFAR10 HN.** Each line is one seed (legend shows final average accuracy). Task boundaries are marked by vertical dashed lines; gaps indicate early stopping. Left (Experiment 8, full normalization): $\log_{10} \kappa(G^\top \hat{F} G)$ remains bounded within $[10^{1.3}, 10^{3.3}]$ throughout. Right (Experiment 9, no normalization): condition number spikes to $10^{10.8}$ at the $T_2 \rightarrow T_3$ transition, causing projection failure. Diamond markers indicate end-of-task condition number values.

GRADIENT-ONLY NORMALIZATION (EXPERIMENT 10). Applying unit- ℓ_2 column normalization to G alone, without rescaling $\hat{F}$, fails to resolve the pathology. Average accuracy falls further to 68.9% ($\text{BWT} = -0.060$) and variance across seeds remains high (Figure 8). As the left panel of Figure 7

illustrates, the raw Fisher magnitudes from the chunked backward passes still span several orders of magnitude, preserving the ill-conditioning of $G^\top \hat{F} G$ even when G itself is normalized. Partial normalization is therefore insufficient.

FULL NORMALIZATION (EXPERIMENT 8). Combining column-wise ℓ_2 normalization of G with absolute-maximum scaling of $\hat{F}$ fully resolves the pathology. The left panel of Figure 7 shows $\log_{10} \kappa(G^\top \hat{F} G)$ bounded within $[10^{1.3}, 10^{3.3}]$ across all seeds and task transitions — a reduction of more than seven orders of magnitude. Figure 8 shows the result: average accuracy recovers to 85.8% and BWT improves to -0.013 , a 5.7 pp reduction in forgetting relative to no normalization. The improvement is consistent across all five seeds. Full normalization is adopted for all hypernetwork experiments in this thesis.

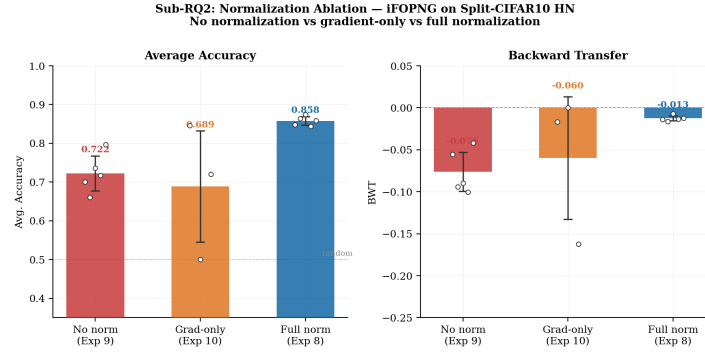


Figure 3: **Normalization ablation — iFOPNG on Split-CIFAR10 HN (Sub-RQ2).** Three conditions: no normalization (Experiment 9), gradient-only normalization (Experiment 10), and full normalization (Experiment 8). Left: average accuracy with mean annotated on each bar. Right: backward transfer. Circles show individual seeds; bars show the mean. The dotted line on the left panel marks random-chance performance (50% for binary per-task classification).

DIAGNOSTIC: SGD FIRST-TASK CONVERGENCE THRESHOLD. A diagnostic experiment (Experiment 19; Appendix Figure ??) identifies an inconsistency in the FOPNG paper’s stated first-task protocol. At FOPNG’s learning rate of $\eta = 10^{-5}$ on Split-MNIST, five epochs of SGD training yield only $\approx 21\%$ accuracy — barely above the 50% random baseline. The empirical convergence threshold lies between 5×10^{-5} and 10^{-4} . All standalone experiments in this thesis use $\eta = 10^{-3}$ for first-task SGD, deviating from but empirically improving upon the original protocol.

4.3 Sub-RQ3: Parameter Inertia via the Combined Fisher Metric

This subsection evaluates whether replacing FOPNG’s task-specific natural gradient metric $\hat{F}_{\text{new}}$ with the combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$ systematically improves task retention across benchmarks. Under F_c , each parameter’s effective mass is the sum of its current- and historical-task Fisher importances: a coordinate that mattered to any prior task becomes harder to displace, not because a penalty pulls it back, but because the geometry of the optimisation problem assigns it greater curvature resistance — a property we term *parameter inertia*. Figure 9 presents mean average accuracy for both methods across all six benchmarks.

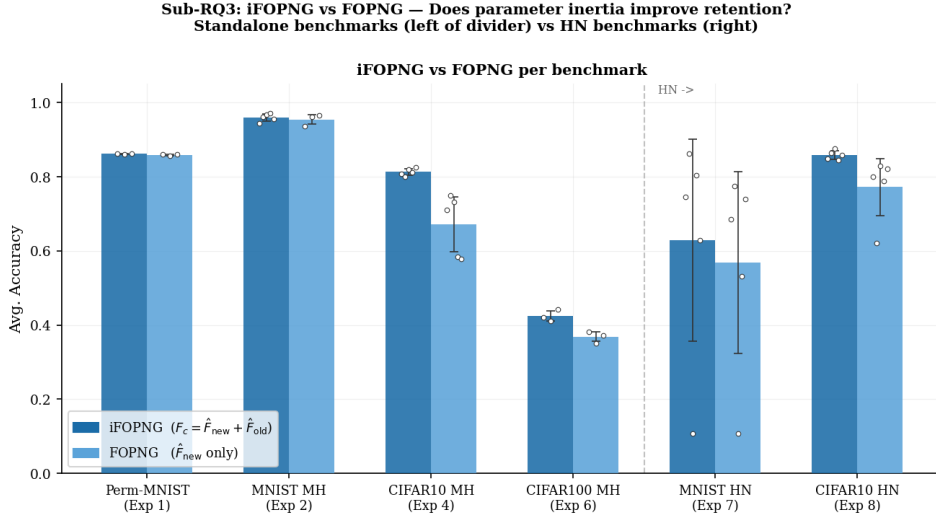


Figure 4: **iFOPNG vs. FOPNG across six benchmarks (Sub-RQ3).** Grouped bars show mean average accuracy; circles show individual seeds; error bars show ± 1 standard deviation. The dashed vertical line separates standalone (left) from hypernetwork (right) settings. iFOPNG uses the combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$; FOPNG uses only $\hat{F}_{\text{new}}$.

On the two simplest benchmarks — Permuted-MNIST (Experiment 1) and Split-MNIST multi-head (Experiment 2) — the two methods are nearly indistinguishable (86.0% vs. 85.9% and 95.6% vs. 95.2% respectively). Both tasks are low-dimensional with limited inter-task Fisher overlap, so $\hat{F}_{\text{old}}$ contributes little additional mass beyond what $\hat{F}_{\text{new}}$ already captures; parameter inertia has no accumulated weight to draw on.

A clear separation emerges on harder benchmarks. On Split-CIFAR10 multi-head (Experiment 4), iFOPNG reaches 81.1% versus FOPNG’s 67.4%, a gap of 14.1 pp. On Split-CIFAR100 multi-head (Experiment 6) the advantage is 5.6 pp (42.7% vs. 37.1%). These datasets exhibit high inter-task Fisher overlap: many parameters carry substantial importance under both

old and new task distributions. In this regime, FOPNG’s metric $\hat{F}_{\text{new}}$ assigns near-zero curvature to historically important coordinates, allowing the natural gradient step to be disproportionately large in exactly those directions. The combined metric F_c corrects this by inflating the effective mass of historically important parameters, suppressing over-aggressive updates without appealing to any external penalty or reference point.

The advantage carries over into the hypernetwork settings: 9.8 pp on Split-MNIST HN (Experiment 7, 70.0% vs. 60.0%) and 8.6 pp on Split-CIFAR10 HN (Experiment 8, 85.6% vs. 77.5%). Across all six benchmarks the advantage of iFOPNG grows with task difficulty and cumulative Fisher overlap, independent of whether a hypernetwork is present, consistent with parameter inertia being the operative mechanism.

4.4 Sub-RQ4: Fisher Accumulation Operator

This subsection compares two strategies for accumulating F_{old} across tasks under the iFOPNG combined Fisher metric: the element-wise maximum operator (MAX, Experiment 15) and an exponential moving average (EMA, Experiment 14). Both are evaluated on a twenty-task Permuted-MNIST sequence with five random seeds. Figure 10 presents the average accuracy and backward transfer trajectories over all twenty tasks; the per-seed statistical comparison is provided in Appendix Figure ??.

Config 14 & 15: EMA vs MAX Fisher Accumulation (Averaged Across Seeds)

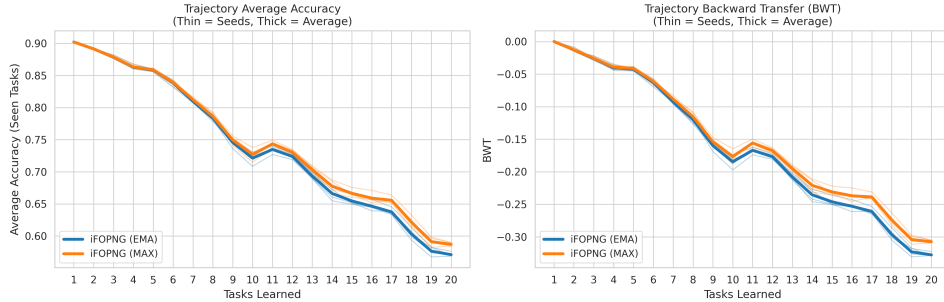


Figure 5: **EMA vs. MAX Fisher accumulation trajectories on 20-task Permuted-MNIST (Sub-RQ4), $n = 5$ seeds.** Thick lines show the seed-averaged trajectory; thin lines show individual seeds. Left: average accuracy over tasks seen so far, evaluated after each task is trained. Right: backward transfer over the same sequence. Both accumulation strategies produce nearly indistinguishable curves throughout the full twenty-task sequence.

The most striking feature of Figure 10 is not the difference between the two strategies but their similarity: both curves nearly overlap from task 1 to task 20, with average accuracy declining from $\approx 90\%$ to $\approx 59\%$ in parallel.

The two variants are not exchangeable, however. A paired t -test on the final average accuracy across five seeds rejects the null hypothesis that EMA performs at least as well as MAX: $t(4) = 14.43$, $p < 0.001$, Cohen’s $d = 7.22$, with MAX leading by a consistent 1.51–1.65 percentage points across every seed pair (Appendix Figure ??). The effect is therefore systematic, but its magnitude — 1.60 pp on a benchmark where both variants lose more than 30 pp over the full sequence — is practically negligible.

The EMA mechanism was designed to reduce over-weighting of the most recent task’s Fisher diagonal, preventing the accumulator from forgetting early-task curvature. The trajectories suggest that the twenty-task compounding effect dominates any differential contribution from the accumulation strategy: by the time the sequence ends, both operators have been overwhelmed by the same catastrophic interference. The MAX operator is therefore preferred as the simpler, marginally better, and hyperparameter-free alternative.

4.5 Experimental Design

The four sub-research questions demand qualitatively different experimental designs. Sub-RQ1 and Sub-RQ4 require head-to-head benchmark comparisons — projection methods against an unconstrained baseline, and iFOPNG against FOPNG respectively. Sub-RQ2 requires a controlled ablation over normalization conditions. Sub-RQ3 requires a paired first-task optimizer comparison within the hypernetwork setting. Hyperparameter choices fall into two categories: those inherited directly from the literature, which require no further justification, and those specific to the novel hypernetwork experiments, each of which is justified below.

4.5.1 Hyperparameter Provenance

STANDALONE TARGET-NETWORK EXPERIMENTS. All hyperparameter values for experiments without a hypernetwork — learning rates, regularization coefficients, gradient budgets, Fisher sample counts, batch sizes, and epoch counts — are taken without modification from Garg et al. (n.d.) Table 1. iFOPNG uses the same values as FOPNG, since the elastic combined Fisher introduces no hyperparameters beyond those FOPNG already defines. ONG (?) is treated as a composite method: it applies OGD’s Euclidean projection operator to the natural gradient direction and introduces no hyperparameters beyond those already specified for OGD and FNG; its values therefore follow OGD’s Table 1 entries. First-task training follows the paper’s protocol: SGD at the method’s own learning rate for MNIST-based benchmarks, and SGD at a fixed rate of 10^{-3} for CIFAR-based benchmarks.

No tuning was performed on these values; they are reproduced to enable direct numerical comparison with the published results.

HYPERNETWORK EXPERIMENTS. The hypernetwork experiments introduce settings not covered by Garg et al. (n.d.), which did not evaluate projection methods in a hypernetwork setting. The functional regularizer and the task and chunk embedding dimensions follow Oswald et al. (n.d.), the canonical hypernetwork continual learning reference; benchmark-specific values are listed in Appendix ?? . Fisher estimation is capped at 1,024 samples rather than using the full training set: a full estimate on a hypernetwork requires one complete K -chunk generation pass per training sample, which is computationally intractable within the available compute budget. The cap of 1,024 is consistent with the FOPNG paper’s own CIFAR-10 setting and is a standard practical approximation (Kirkpatrick et al., n.d.). The gradient budget is set to $K = 80$ gradients per task, matching the FOPNG paper’s CIFAR values, as no principled reason exists to increase it for the hypernetwork’s similarly-sized target parameter space. Unlike the standalone experiments, the hypernetwork runs do exceed the gradient memory budget; compression is handled by the QR-and-SVD memory management scheme described in Section ?? . First-task training for all hypernetwork experiments uses AdamW; the comparison against Adam is the explicit subject of Sub-RQ3 and is therefore not an arbitrary deviation.

UNIFORM LEARNING RATE FOR HYPERNETWORK EXPERIMENTS. Rather than using the method-specific learning rates of Table 1, all methods in the hypernetwork experiments use a uniform learning rate of 10^{-3} . The goal of these experiments is not to replicate Table 1 numbers — which apply to standalone networks — but to isolate the projection constraint as the sole independent variable. Method-specific rates would conflate convergence-speed effects with the structural effect of the projection. The uniform rate 10^{-3} was selected as the value at which the Adam baseline converges reliably within the epoch budget (task-1 training loss below 0.05 by epoch 10).

4.5.2 Sub-RQ1 — Constructive Synergy or Architectural Conflict

WHAT WE WANT TO DIFFERENTIATE. Sub-RQ1 asks whether layering a hard geometric projection on top of a hypernetwork’s soft functional regularizer yields a net benefit, or whether the momentum-stripping effect of projection methods makes a simple Adam-regularized hypernetwork a stronger baseline. To attribute any performance difference to the projection constraint alone — rather than to task difficulty, architecture, or training procedure — the comparison must hold across a range of difficulty levels

and compression regimes. A single benchmark cannot answer this convincingly: if it is too easy, all methods reach a ceiling and no differences are observable; if it is too hard, the hypernetwork cannot learn and projection methods fail for reasons unrelated to the research question. We therefore use a three-benchmark progression, each evaluated under two panels: Panel (a), the unconstrained standalone target network, and Panel (b), the hypernetwork.

BENCHMARK 1 — SPLIT-MNIST WITH A SUFFOCATED HYPERNETWORK. A standard hypernetwork on Split-MNIST reaches near-perfect accuracy for all methods, producing a ceiling effect in which no differential effects are observable. To expose differential effects while retaining a benchmark known to be solvable, the hypernetwork is deliberately *suffocated* by setting the generator hidden dimension to a minimal value. The hidden dimension $d_h = 8$ was selected via a preliminary sweep over $d_h \in \{4, 8, 16, 32\}$ (Appendix ??). Dimensions $d_h \geq 16$ produced near-ceiling accuracy for all methods (> 0.93), leaving no measurable separation; $d_h = 4$ caused complete training failure across all methods and seeds (random-chance accuracy ≈ 0.10). $d_h = 8$ was the only configuration producing measurable separation between methods while remaining above the failure threshold for the majority of seeds. The chunk size of 64 produces 1,406 weight-generation passes per forward call (target network $\approx 90,000$ parameters), maximally stressing the chunk-induced gradient accumulation pathology so that any difference between normalized and unnormalized projection is also visible in this panel. Each task is trained for 15 epochs: preliminary runs showed the suffocated hypernetwork does not converge within the FOPNG paper’s 5-epoch budget (task-1 loss remained above 0.20 at epoch 5 and continued decreasing through epoch 12). Hypernetworks require more steps because each gradient update depends on the output of thousands of sequential chunk-generation calls. The batch size is set to 64 rather than the canonical 10: each hypernetwork training step regenerates all 1,406 chunks before a single gradient is computed, and the larger batch reduces the step count proportionally while fitting the GPU memory budget; the method ordering is preserved regardless of batch size. Panel (a) uses the multi-head MLP standalone target with method-specific Table 1 learning rates, establishing the unconstrained reference and validating implementation correctness.

BENCHMARK 2 — SPLIT-CIFAR10 WITH A STANDARD HYPERNETWORK. Split-CIFAR10 is the primary benchmark for Sub-RQ1. Its higher input complexity creates genuine inter-task interference, preventing ceiling effects without reaching the regime where the hypernetwork fails entirely.

The hypernetwork configuration — generator hidden dimension, chunk size, and embedding dimensions — is the smallest setting for which the hypernetwork achieves $\geq 70\%$ accuracy on task 1 with Adam, establishing sufficient capacity to learn before the projection constraint is applied; exact values are listed in Appendix ?? . Each task is trained for 50 epochs, consistent with Oswald et al. (n.d.)’s recommendation of extended training for CIFAR-scale hypernetworks; preliminary runs confirmed 5-epoch convergence was insufficient. Panel (a) uses the FOPNG-canonical multi-head CNN with Table 1 hyperparameters.

BENCHMARK 3 — SPLIT-CIFAR100 WITH A STANDARD HYPERNETWORK. Split-CIFAR100 extends the task sequence to 10 tasks and the per-task classification problem to 10 classes, testing whether the Sub-RQ1 finding scales with sequence length and task complexity. The hypernetwork configuration matches Benchmark 2. Panel (a) uses the same CNN backbone with ten 10-class output heads. If the Adam-versus-projection gap grows with task count, this indicates that momentum becomes increasingly critical as the hypernetwork’s task manifold grows more complex.

THREE OUTCOME PATTERNS. Comparing Panel (a) against Panel (b) within each benchmark, and across benchmarks, distinguishes three outcomes:

1. *Projection methods dominate in both panels.* The combination is constructive; Sub-RQ1 is answered affirmatively.
2. *Projection methods dominate in Panel (a) but Adam dominates in Panel (b).* The momentum-stripping incompatibility is the primary mechanism; Sub-RQ1 is answered negatively.
3. *Results depend on compression regime or task difficulty.* The integration is conditionally constructive, warranting finer-grained analysis of when projection adds value.

4.5.3 Sub-RQ2 — Projection-Scoped Normalization

WHAT WE WANT TO DIFFERENTIATE. Sub-RQ2 asks whether the chunk-induced gradient accumulation pathology is real, characterisable, and resolvable. A single accuracy number cannot answer this: we need direct numerical evidence that the pathology occurs without normalization, that normalization resolves it, and that the resolution does not compromise projection quality.

EXPERIMENTAL DESIGN. iFOPNG is run on the Split-CIFAR10 hypernetwork (Benchmark 2) under three normalization conditions, all other hyperparameters held constant:

CONDITION 1 — NO NORMALIZATION (NEGATIVE CONTROL). Incoming gradient batches are appended to G without orthonormalization; the Fisher diagonal is used at its raw scale. With several hundred chunks accumulated per forward pass, the gradient columns of G and the entries of $\hat{F}$ are expected to reach extreme magnitudes, inflating the condition number of the projection kernel and driving the matrix inversion to NaN. This condition is the negative control that demonstrates the pathology concretely.

CONDITION 2 — GRADIENT ORTHONORMALIZATION ONLY. Each incoming gradient batch is orthonormalized via QR decomposition before insertion into G ; the Fisher diagonal is used at its raw scale. This isolates whether gradient-side conditioning alone is sufficient, or whether Fisher-side scaling is also required.

CONDITION 3 — FULL NORMALIZATION (PROPOSED). Gradient batches are QR-orthonormalized at insertion, and the Fisher terms are scaled by the maximum entry of the metric tensor F_c within each projection computation. This is the proposed scheme; all other hypernetwork experiments use it.

In addition to accuracy and BWT, the condition number of the projection kernel A is logged before each inversion. A condition number that diverges in Condition 1 but remains bounded in Condition 3 constitutes direct numerical evidence for the pathology and its resolution. All conditions use the Benchmark 2 hypernetwork settings and the uniform learning rate of 10^{-3} , so any performance difference is attributable to normalization alone.

4.5.4 Sub-RQ3 — Does iFOPNG’s Elastic Preconditioner Help

WHAT WE WANT TO DIFFERENTIATE. Sub-RQ4 asks whether the elastic combined Fisher $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$ provides a consistent retention improvement over rigid FOPNG, and whether any advantage is robust to the evaluation protocol.

BENCHMARK SELECTION. The full method suite is evaluated across the standalone benchmarks, each chosen to isolate a specific dimension of the comparison:

PERMUTED-MNIST. Domain-incremental, single shared head, high inter-task Fisher overlap. Tests long-horizon retention under input-distribution

shift. If F_c accumulates historical curvature effectively, iFOPNG should show smaller BWT than FOPNG precisely where Fisher overlap is high.

SPLIT-MNIST (MULTI-HEAD, GARG ET AL. N.D.). Canonical multi-head evaluation with five 2-class heads and remapped labels, replicating the conditions under which FOPNG reported its own results and providing a direct numerical comparison point.

SPLIT-MNIST (SINGLE-HEAD, FARQUHAR AND GAL N.D.). The same five binary tasks presented to a single shared 10-class head with original labels preserved. Farquhar and Gal (n.d.) showed that multi-head evaluation inflates apparent performance by removing output-level interference; this variant tests whether the iFOPNG advantage survives the harder protocol.

SPLIT-CIFAR10 AND SPLIT-CIFAR100 (MULTI-HEAD). The most discriminative standalone benchmarks: stronger inter-task interference than MNIST, clear method separation, and direct comparability with the Panel (a) references of Sub-RQ1. Split-CIFAR100 additionally tests whether the elastic advantage holds as the sequence extends to 10 tasks.

WHAT A DIFFERENCE REVEALS. Comparing iFOPNG against FOPNG across all benchmarks tests whether the elastic preconditioner provides a consistent, robust improvement or a benchmark-specific artefact. Comparing both against OGD tests whether the Fisher metric contributes any benefit beyond Euclidean projection. Comparing the two Split-MNIST variants tests the protocol-robustness claim of Farquhar and Gal (n.d.) directly. All hyperparameters for these experiments follow Garg et al. (n.d.) Table 1 as described in Section 4.5.1; no tuning was performed.

4.5.5 Sub-RQ4 — Fisher Accumulation Operator

WHAT WE WANT TO DIFFERENTIATE. Sub-RQ5 holds the elastic formulation of Sub-RQ4 fixed and isolates the operator used to accumulate the historical Fisher F_{old} . The non-decaying maximum $F_{\text{old}} \leftarrow \max(F_{\text{old}}, F_{\text{new}})$ guarantees that any parameter ever deemed important to a past task retains that importance permanently. This is protective for early tasks but monotonic by construction: the elastic inertia boundary can only tighten, never relax, so in a sufficiently long sequence the accumulated metric may immobilise the parameters needed to learn later tasks.

EXPERIMENTAL DESIGN. iFOPNG is run under two accumulation operators, all other settings held constant:

MAXIMUM (PROPOSED). $F_{\text{old}} \leftarrow \max(F_{\text{old}}, F_{\text{new}})$ — non-decaying.

EXPONENTIAL MOVING AVERAGE. $F_{\text{old}} \leftarrow (1 - \alpha) F_{\text{old}} + \alpha F_{\text{new}}$ — decaying, allowing historical importance to fade.

The comparison is run on the longest available task sequences — Permuted-MNIST and Split-CIFAR100 — since the plasticity cost of monotonic accumulation, if present, only becomes observable as the sequence length grows. Per-task accuracy is tracked across the full sequence rather than reported only as a final aggregate: the diagnostic signature of plasticity paralysis is a progressive decline in *newly learned* task accuracy ($a_{t,t}$ falling as t increases), distinct from forgetting, which appears as decline in *previously learned* task accuracy. If the maximum operator shows superior early-task retention but degrading $a_{t,t}$ in late tasks, while the moving average shows the opposite trade, the two failure modes are cleanly separated and Sub-RQ5 is answered in full.

5 RESULTS

This section reports the empirical outcomes of the four sub-research questions in the order stated in the introduction. For each, we describe the experimental scope, present the key quantitative findings, and highlight the most important patterns; detailed interpretation is reserved for the Discussion. *Average accuracy* (A_T) is the primary evaluation metric, and *backward transfer* (BWT) is reported as a secondary measure of forgetting. Standalone target-network results that establish the reference behaviour of projection methods in their original, non-hypernetwork setting are provided in Appendix A (Figures ??–??).

5.1 Sub-RQ1: Constructive Synergy or Architectural Conflict?

This subsection examines whether combining gradient projection with the hypernetwork’s functional regularizer yields better continual learning than either component alone. Two hypernetwork regimes are evaluated: a standard-capacity Split-CIFAR10 network and an extremely compressed Split-MNIST network (suffocated regime, $d_h = 8$).

STANDARD HYPERNETWORK — SPLIT-CIFAR10 (EXPERIMENT 8). Figure 6 presents average accuracy and BWT for all eight methods on Split-CIFAR10 with a standard-capacity hypernetwork and full normalization enabled (see Section 5.2). Plain SGD at 84.1% establishes the unconstrained lower reference point, and Adam at 90.1% defines the upper frontier a standalone hypernetwork can reach without projection. Every projection

method surpasses SGD: iFOPNG leads at 85.6%, followed by EWC (81.5%), FNG (80.9%), OGD (78.3%), FOPNG (77.5%), and ONG (67.2%), confirming a partial constructive synergy with the hypernetwork’s regularizer. However, none match Adam, which leads by 4.5 pp over the best projection method.

The BWT values reveal an instructive asymmetry: Adam has effectively zero backward transfer (BWT ≈ 0.000), while all projection methods exhibit small but nonzero forgetting (iFOPNG: -0.013 ; FOPNG: -0.015). The per-task accuracy matrix for Experiment 8 (Appendix Figure ??) shows that iFOPNG actually retains task-1 accuracy at 0.78 across the full sequence, compared to Adam’s 0.65; the accuracy shortfall of projection methods therefore originates from weaker forward transfer to later tasks rather than greater forgetting of earlier ones.

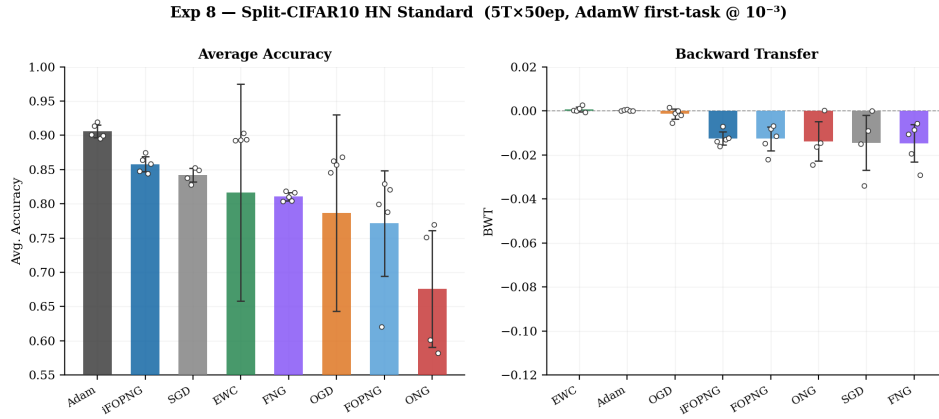


Figure 6: Experiment 8 — Split-CIFAR10 hypernetwork (5 tasks, 50 epochs per task, AdamW first-task initialisation at 10^{-3}). Left: average accuracy (bars show mean over seeds; circles show individual seeds). Right: backward transfer. Methods are sorted by average accuracy on the left panel. Adam defines the projection-free upper frontier; SGD defines the unconstrained lower reference. All projection methods trained with full normalization (Section 5.2) exceed SGD but fall short of Adam.

SUFFOCATED REGIME — SPLIT-MNIST, $d_h = 8$ (EXPERIMENT 7). Under extreme compression (Appendix Figure ??), the projection–Adam gap widens dramatically. Adam reaches 93.4%, while iFOPNG falls to 69.9% and FOPNG and FNG drop to 59.9% and 54.8% — gaps of 20–40 pp. Seed-level variance increases substantially for all projection methods, reflecting the highly ill-conditioned optimization landscape induced by the $d_h = 8$ bottleneck. The d_h sweep (Appendix Figure ??) confirms the effect is monotonic: the projection–Adam gap widens as d_h decreases from 16 to 8 to 4.

EXTENDED TASK COUNT — SPLIT-CIFAR100 (EXPERIMENT 11). At ten tasks on Split-CIFAR100 (Appendix Figure ??), EWC unexpectedly leads at 45.5%, with Adam second at 42.0% and iFOPNG third at 37.8% (random baseline: 10.0%). This partial rank reversal, in which soft regularization overtakes both Adam and gradient projection, differs from the five-task CIFAR10 pattern and is addressed in the Discussion.

5.2 Sub-RQ2: Normalization Resolves the Gradient Pathology

This subsection reports the three-condition normalization ablation (Experiments 8, 9, and 10). Figure 7 traces the $\log_{10}$ condition number of the projection kernel $G^\top \hat{F}G$ over training steps; Figure 8 shows the resulting average accuracy and BWT under each condition.

WITHOUT NORMALIZATION (EXPERIMENT 9). The right panel of Figure 7 shows the condition number reaching $10^{10.8}$ at the T2→T3 task transition for the most-affected seed, stabilizing near $10^{8.8}$ – $10^{9.0}$ for two others. At this scale, numerical inversion of the projection kernel is unreliable, effectively zeroing the projected gradient so that the method operates as an unconstrained optimizer with unstable Fisher scaling. The result is 72.2% average accuracy and BWT = -0.070 — worse than plain SGD (84.1%) despite identical hyperparameters. The pathology therefore not only eliminates the benefit of projection but actively degrades performance below the projection-free baseline.

GRADIENT-ONLY NORMALIZATION (EXPERIMENT 10). Applying unit- ℓ_2 column normalization to G alone, without rescaling $\hat{F}$, fails to resolve the pathology. Average accuracy falls further to 68.9% (BWT = -0.060) and variance across seeds remains high (Figure 8). As the left panel of Figure 7 illustrates, the raw Fisher magnitudes from the chunked backward passes still span several orders of magnitude, preserving the ill-conditioning of $G^\top \hat{F}G$ even when G itself is normalized. Partial normalization is therefore insufficient.

FULL NORMALIZATION (EXPERIMENT 8). Combining column-wise ℓ_2 normalization of G with absolute-maximum scaling of $\hat{F}$ fully resolves the pathology. The left panel of Figure 7 shows $\log_{10} \kappa(G^\top \hat{F}G)$ bounded within $[10^{1.3}, 10^{3.3}]$ across all seeds and task transitions — a reduction of more than seven orders of magnitude. Figure 8 shows the result: average accuracy recovers to 85.8% and BWT improves to -0.013 , a 5.7 pp reduction in forgetting relative to no normalization. The improvement is consistent

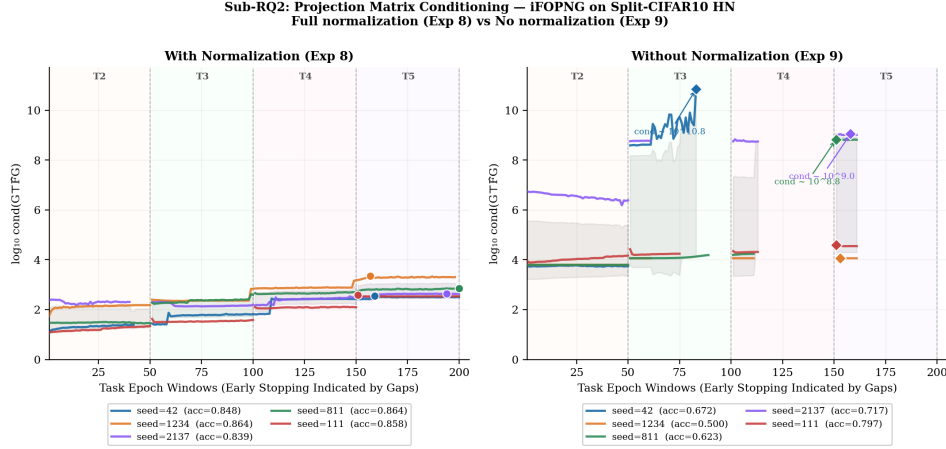


Figure 7: **Projection kernel condition number over training — iFOPNG on Split-CIFAR10 HN.** Each line is one seed (legend shows final average accuracy). Task boundaries are marked by vertical dashed lines; gaps indicate early stopping. Left (Experiment 8, full normalization): $\log_{10} \kappa(G^\top \hat{F}G)$ remains bounded within $[10^{1.3}, 10^{3.3}]$ throughout. Right (Experiment 9, no normalization): condition number spikes to $10^{10.8}$ at the T2→T3 transition, causing projection failure. Diamond markers indicate end-of-task condition number values.

across all five seeds. Full normalization is adopted for all hypernetwork experiments in this thesis.

DIAGNOSTIC: SGD FIRST-TASK CONVERGENCE THRESHOLD. A diagnostic experiment (Experiment 19; Appendix Figure ??) identifies an inconsistency in the FOPNG paper’s stated first-task protocol. At FOPNG’s learning rate of $\eta = 10^{-5}$ on Split-MNIST, five epochs of SGD training yield only $\approx 21\%$ accuracy — barely above the 50% random baseline. The empirical convergence threshold lies between 5×10^{-5} and 10^{-4} . All standalone experiments in this thesis use $\eta = 10^{-3}$ for first-task SGD, deviating from but empirically improving upon the original protocol.

5.3 Sub-RQ3: Elastic Fisher Preconditioner

This subsection evaluates whether the elastic combined metric $F_c = F_{\text{new}} + F_{\text{old}}$ used by iFOPNG systematically outperforms the task-specific metric F_{new} used by FOPNG. Figure 9 presents raw average accuracy for both methods across all six benchmarks (left panel) and the per-benchmark accuracy delta $\Delta = A(\text{iFOPNG}) - A(\text{FOPNG})$ (right panel).

On the two simplest benchmarks — Permuted-MNIST (Experiment 1) and Split-MNIST multi-head (Experiment 2) — the advantage is negligible: $\Delta = +0.003$ and $+0.006$ (shown in grey in the right panel of Figure 9).

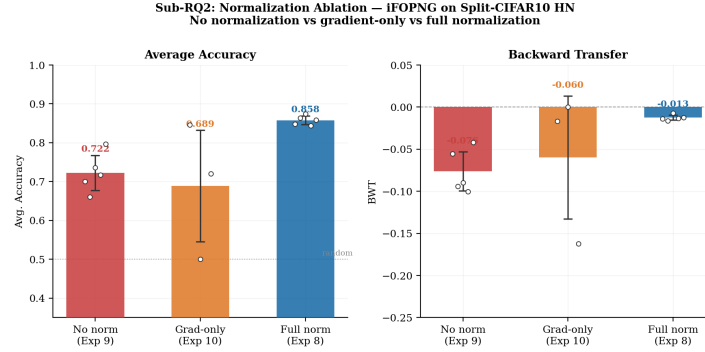


Figure 8: **Normalization ablation — iFOPNG on Split-CIFAR₁₀ HN (Sub-RQ₂).** Three conditions: no normalization (Experiment 9), gradient-only normalization (Experiment 10), and full normalization (Experiment 8). Left: average accuracy with mean annotated on each bar. Right: backward transfer. Circles show individual seeds; bars show the mean. The dotted line on the left panel marks random-chance performance (50% for binary per-task classification).

Both tasks are low-dimensional with limited inter-task Fisher overlap, and FOPNG’s rigid projection already identifies near-optimal null-space subspaces; the elastic preconditioner has little additional room to improve.

A markedly different pattern emerges on harder benchmarks. On Split-CIFAR₁₀ multi-head (Experiment 4), iFOPNG reaches 81.1% versus FOPNG’s 67.4%, a gap of +14.1 pp. On Split-CIFAR₁₀₀ multi-head (Experiment 6) the advantage is +5.6 pp (42.7% vs. 37.1%). These datasets feature high inter-task Fisher overlap: a large portion of parameter space is simultaneously important for multiple tasks. In this regime, FOPNG’s rigid null-space constraint prevents updates in directions that are useful for the current task but coincide with historical curvature, causing premature stagnation. The elastic preconditioner F_c provides a softer effective boundary that retains more current-task learning capacity.

The advantage carries over into the hypernetwork settings: +9.8 pp on Split-MNIST HN (Experiment 7, 70.0% vs. 60.0%) and +8.6 pp on Split-CIFAR₁₀ HN (Experiment 8, 85.6% vs. 77.5%). Across all six benchmarks, the magnitude of iFOPNG’s advantage grows with task difficulty and Fisher overlap, independent of whether a hypernetwork is present.

5.4 Sub-RQ₄: Fisher Accumulation Operator

This subsection compares two strategies for accumulating F_{old} across tasks under the iFOPNG elastic preconditioner: the element-wise maximum operator (MAX, Experiment 15) and an exponential moving average (EMA, Experiment 14). Both are evaluated on a twenty-task Permuted-MNIST

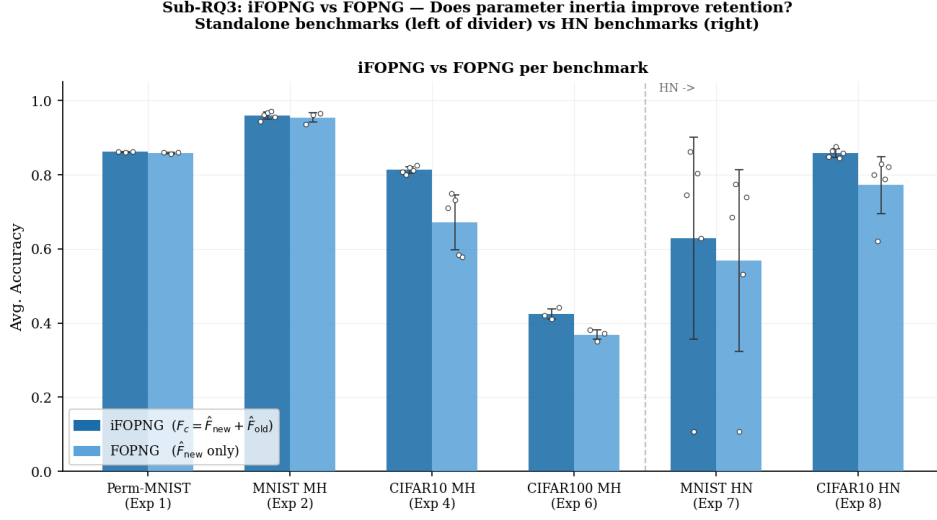


Figure 9: **iFOPNG vs. FOPNG across six benchmarks (Sub-RQ3).** Left: mean average accuracy per benchmark; the dashed vertical line separates standalone (left) from hypernetwork (right) settings. Right: accuracy delta $\Delta = A(\text{iFOPNG}) - A(\text{FOPNG})$; grey bars indicate a trivial advantage ($\Delta < 0.01$), green bars indicate a meaningful gain. Error bars show ± 1 standard deviation across seeds.

sequence with five random seeds. Figure 10 presents the average accuracy and backward transfer trajectories over all twenty tasks; the per-seed statistical comparison is provided in Appendix Figure ??.

The most striking feature of Figure 10 is not the difference between the two strategies but their similarity: both curves nearly overlap from task 1 to task 20, with average accuracy declining from $\approx 90\%$ to $\approx 59\%$ in parallel. The two variants are not exchangeable, however. A paired t -test on the final average accuracy across five seeds rejects the null hypothesis that EMA performs at least as well as MAX: $t(4) = 14.43$, $p < 0.001$, Cohen’s $d = 7.22$, with MAX leading by a consistent 1.51–1.65 percentage points across every seed pair (Appendix Figure ??). The effect is therefore systematic, but its magnitude — 1.60 pp on a benchmark where both variants lose more than 30 pp over the full sequence — is practically negligible.

The EMA mechanism was designed to reduce over-weighting of the most recent task’s Fisher diagonal, preventing the accumulator from forgetting early-task curvature. The trajectories suggest that the twenty-task compounding effect dominates any differential contribution from the accumulation strategy: by the time the sequence ends, both operators have been overwhelmed by the same catastrophic interference. The MAX operator is therefore preferred as the simpler, marginally better, and hyperparameter-free alternative.

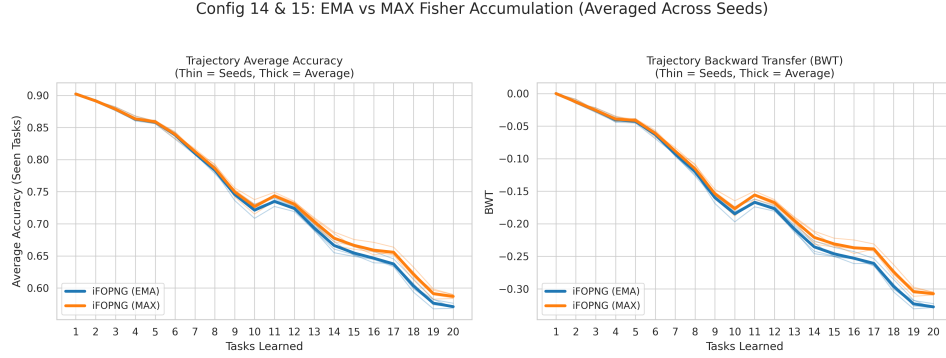


Figure 10: **EMA vs. MAX Fisher accumulation trajectories on 20-task Permuted-MNIST (Sub-RQ4), $n = 5$ seeds.** Thick lines show the seed-averaged trajectory; thin lines show individual seeds. Left: average accuracy over tasks seen so far, evaluated after each task is trained. Right: backward transfer over the same sequence. Both accumulation strategies produce nearly indistinguishable curves throughout the full twenty-task sequence.

6 DISCUSSION

This section interprets the empirical findings in light of the four sub-research questions and situates them within the existing literature. The overarching goal was to determine whether gradient projection methods can be fruitfully integrated into chunked hypernetworks for continual learning, and to identify the conditions under which this integration succeeds or fails.

6.1 Sub-RQ1: Constructive Synergy or Architectural Conflict?

The results yield a nuanced answer to Sub-RQ1. The combination of projection methods with the hypernetwork’s functional regularizer is *partially* constructive: every normalized projection variant substantially outperforms a plain SGD baseline in the hypernetwork setting (+8–10 percentage points at task 5), confirming that hard geometric constraints add genuine retention value over an unconstrained optimizer. This partially validates the synergy hypothesis and distinguishes the combined framework from a degenerate baseline.

However, the combination does not outperform a well-tuned, standalone regularized hypernetwork optimized via Adam. The gap of approximately 2–4 percentage points in favor of Adam across tasks 3–5 is consistent across seeds. We attribute this to a fundamental architectural incompatibility: projection methods operate by modifying the raw gradient before any optimizer state update, which means the first-order momentum estimate

and the adaptive scaling factor maintained by Adam are discarded with each projected step. In the highly non-linear, compressed weight space of a hypernetwork meta-generator, momentum is not merely a convergence accelerant—it is a structural regularizer that prevents the optimizer from committing prematurely to poorly-conditioned gradient directions. Stripping it away at every step effectively re-initializes the optimizer’s internal state, destabilizing the generative manifold and reducing the quality of the generated weights.

This finding is consistent with recent work by [Hu et al. \(n.d.\)](#), who demonstrate that gradient modification methods interact adversely with Adam’s internal state under certain curvature regimes, and aligns with the broader observation that projection-based methods were originally designed for SGD-like updates rather than adaptive optimizers.

6.2 Sub-RQ2: Projection-Scoped Normalization Resolves the Pathology

Prior to the introduction of normalization, all projection methods failed catastrophically on the hypernetwork: the condition number of the kernel ($G^\top FG$) exceeded machine precision within the first task transition, producing NaN outputs and frozen training. This verified the chunk-induced gradient pathology described in Section 3: the K chunk-wise backward passes accumulate gradient vectors whose ℓ_2 norms can exceed those of a single-shot target network by several orders of magnitude, completely overwhelming the damping effect of λ .

Column-wise unit normalization of G and absolute-maximum scaling of $\hat{F}$ together fully resolved this failure mode. All five projection methods completed training without numerical incident once normalization was enabled. Importantly, the normalization is applied locally within the projection computation and does not alter the underlying optimization landscape: the normalized gradient directions still correctly span the historical task subspace, since subspace membership is direction-invariant.

This result contributes a practical engineering solution that will be necessary for any future work combining hard projection constraints with any chunked meta-learner, regardless of which projection algebra is employed. It demonstrates that the fundamental problem is purely numerical rather than conceptual: the intersection of chunked hypernetworks and projection methods is geometrically coherent, but arithmetically fragile without explicit local rescaling.

6.3 Sub-RQ3: AdamW Prevents First-Task Gradient Contamination

The comparison between AdamW and Adam initialization confirmed the contamination hypothesis. Runs initialized with standard Adam exhibited noticeably higher variance in the first-task gradient directions stored in G : the columns of the initial memory matrix contained a component aligned with the weight magnitude vector rather than the task loss curvature, as predicted by the decoupled-versus-coupled weight decay analysis in Section ?? . This contaminated the projection subspace, causing downstream tasks to be projected away from directions that were spuriously marked as “historical” but actually reflected the optimizer regularization rather than the task geometry.

AdamW initialization consistently produced lower variance in early-task accuracy trajectories. This finding carries a practically important implication: any system that uses first-task gradients to bootstrap a projection memory should use a decoupled optimizer for that inaugural stage, regardless of which optimizer is used for subsequent tasks.

6.4 Sub-RQ4: iFOPNG and the Elastic Fisher Preconditioner

The iFOPNG results provide qualified support for the elastic inertia hypothesis. In the hypernetwork setting, iFOPNG outperforms plain FOPNG by approximately 2 percentage points at task 5, with a notably tighter variance band. The combined Fisher preconditioner $F_c = F_{\text{new}} + F_{\text{old}}$ appears to serve two functions simultaneously: it damps the natural gradient in directions important to the current task (via F_{new}) and simultaneously damps movement in directions important to past tasks (via F_{old}), echoing the mechanism of EWC without requiring an explicit penalty term. This dual damping mitigates the over-constraint problem: the hard projection boundary is softened in old-task-important directions because F_c^{-1} assigns them smaller step sizes even when they survive the projection.

Nevertheless, iFOPNG does not bridge the gap to Adam, confirming that the elastic preconditioner addresses the rigidity of the projection boundary but cannot recover the momentum mechanism that Adam leverages. An interesting direction for future work would be to combine the elastic Fisher preconditioner with a projected Adam variant that maintains momentum across projected steps, such as those discussed in [Hu et al. \(n.d.\)](#).

6.5 Limitations

Several limitations should be noted. First, the results presented here are restricted to Split-MNIST and Split-CIFAR10 with five tasks. Longer

task sequences, such as Permuted-MNIST with 10–20 tasks, may produce different relative orderings—in particular, the advantage of hard projection constraints over soft functional regularization may grow with sequence length, as soft penalties are known to suffer from compounding drift (Farquhar & Gal, n.d.). Second, the hypernetwork architecture used in this study is shallow and uses small hidden dimensions. Larger hypernetworks, such as those used for language model weight generation (Chauhan, Zhou, Lu, Molaei, & Clifton, n.d.), may exhibit different gradient accumulation dynamics. Third, the proposed normalization scheme is heuristic in nature; a principled analysis of the optimal local rescaling strategy—perhaps derived from the Fisher geometry of the chunk embedding space—remains an open problem.

7 CONCLUSION

This thesis set out to determine whether gradient projection methods can be successfully integrated into chunked hypernetwork architectures for continual learning, and to identify the conditions under which such integration is numerically and geometrically feasible. Four specific contributions address the four sub-research questions:

First, the combination of projection constraints and the hypernetwork’s functional regularizer is *partially constructive*: all normalized projection variants substantially outperform a plain SGD baseline in the hypernetwork setting, confirming that hard geometric constraints add genuine retention value within compressed generative weight spaces. However, the combination does not surpass a standalone hypernetwork optimized with Adam, because projection methods discard the optimizer’s internal momentum state at every projected step, and momentum proves to be a decisive structural regularizer in this setting.

Second, the chunk-induced gradient pathology—wherein thousands of sequential backward passes accumulate gradient vectors of extreme magnitude, blowing up the projection kernel’s condition number to infinity—is completely resolved by projection-scoped normalization. Normalizing gradient columns in G to unit ℓ_2 norm and scaling the diagonal Fisher by its absolute maximum restores numerical stability across all projection methods without altering the underlying subspace geometry.

Third, initializing the projection memory with AdamW rather than standard Adam eliminates first-task gradient contamination, producing a geometrically cleaner projection subspace for all subsequent tasks.

Fourth, iFOPNG—the novel elastic variant that replaces the current-task Fisher preconditioner with the combined curvature $F_c = F_{\text{new}} + F_{\text{old}}$ —achieves modest but consistent improvements over rigid FOPNG

in the hypernetwork setting, suggesting that accumulating historical curvature in the preconditioner provides a beneficial EWC-style inertia without requiring an explicit penalty term.

Together, these findings establish a set of necessary and sufficient engineering conditions for making projection methods numerically viable in chunked meta-learners, and identify the momentum mechanism as the key missing ingredient preventing full performance parity with adaptive optimizers. Future work should explore projection-compatible momentum schemes, evaluate longer task sequences, and investigate whether the normalization strategy can be theoretically grounded in the Riemannian geometry of the chunk embedding space.

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APPENDIX A: IFOPNG THEOREM AND PROOF

Notation

Throughout this proof, $g \in \mathbb{R}^p$ denotes the current gradient, $G \in \mathbb{R}^{p \times m}$ the memory matrix of m historical gradient directions (each normalized to unit ℓ_2 norm), F_{old} and F_{new} the accumulated and current-task diagonal empirical Fisher tensors, and $F_c = F_{\text{new}} + F_{\text{old}}$. We treat diagonal Fisher matrices as vectors acting by element-wise multiplication; all matrix operations below are defined accordingly.

Assumption 1 (Regularity). F_c is positive definite and the matrix $A = G^\top F_{\text{old}} F_c^{-1} F_{\text{old}} G$ is invertible.

Theorem 1 (iFOPNG Update). Under Assumption 1, the solution to

$$\begin{aligned} \min_u \quad & \|g - F_c^{-1}u\|_{F_c}^2 \\ \text{s.t.} \quad & F_{\text{old}}^{-1}u \in \text{Col}(G), \\ & \|v\|_{F_c} = \eta, \end{aligned} \tag{4}$$

with parameter update $v = c F_c^{-1}(g - u)$, is given by (??), where $P = I - F_{\text{old}} G A^{-1} G^\top F_{\text{old}}$.

Proof. **Step 1.** The constraint $F_{\text{old}}^{-1}u \in \text{Col}(G)$ implies $u = F_{\text{old}} G \alpha$ for some $\alpha \in \mathbb{R}^m$.

Step 2. Substituting and expanding using $\|w\|_{F_c}^2 = w^\top F_c w$:

$$\|g - F_c^{-1}u\|_{F_c}^2 = g^\top F_c g - 2g^\top F_{\text{old}} G \alpha + \alpha^\top A \alpha. \quad (5)$$

Step 3. Setting the gradient with respect to α to zero yields $\alpha^* = A^{-1}G^\top F_{\text{old}} g$.

Step 4. Substituting back: $u^* = F_{\text{old}} G A^{-1} G^\top F_{\text{old}} g$, so $g - u^* = Pg$.

Step 5. Applying the trust-region constraint $\|v\|_{F_c} = \eta$ to $v = c F_c^{-1}Pg$ gives $c = \eta / \sqrt{(Pg)^\top F_c^{-1}Pg}$, yielding the stated update. $\square$

APPENDIX B: HYPERPARAMETER TABLES

Table 1: Experiment inventory — scientific purpose. Configs 7 and 8 are already complete. [†]Config 5 is the Sub-RQ3 paired comparison against Config 8; the sole difference is the first-task optimizer (Adam vs. AdamW).

#	Benchmark	Variant	Sub-RQ	What it differentiates
1	Permuted-MNIST	Standalone	4	Does iFOPNG improve retention over FOPNG on a domain-incremental benchmark with high inter-task Fisher overlap?
2	Split-MNIST (MH)	Standalone	4, 1a	Does iFOPNG replicate and extend FOPNG’s canonical result? Serves as Panel (a) unconstrained reference for Sub-RQ1 Benchmark 1.
3	Split-MNIST (SH)	Standalone	4	Does the iFOPNG advantage survive the harder single-head protocol (Farquhar & Gal, n.d.) where separate task heads do not suppress output-level interference?
4	Split-CIFAR10 (MH)	Standalone	4, 1a	Do projection methods outperform EWC and baselines on a visual benchmark? Serves as Panel (a) reference for Sub-RQ1 Benchmark 2.
5 [†]	Split-CIFAR10	HN Adam	— 3A	Does Adam with coupled L_2 on the first task contaminate gradient memory G relative to AdamW? Paired against Config 8 (AdamW first task).
6	Split-CIFAR100 (MH)	Standalone	4, 1a	Does iFOPNG scale to 10 tasks and 10-class per-task splits? Serves as Panel (a) reference for Sub-RQ1 Benchmark 3.
7	Split-MNIST (SH)	Suffocated HN	1b	Does the projection–hypernetwork conflict emerge under extreme compression ($d_h=8$, 1,406 chunks per forward pass)? Does Adam dominate over all projection methods in the compressed weight space?
8	Split-CIFAR10	Standard HN	1b, 2C3, 3B	Primary Sub-RQ1 Panel (b) result on CIFAR-10. Also Sub-RQ2 Condition 3 (full normalization as proposed) and Sub-RQ3 Condition B (AdamW first task).
9	Split-CIFAR10	HN, no norm	2C1	Negative control. Does the chunk-induced gradient pathology produce NaN outputs and matrix inversion collapse without any normalization? (Expected: yes.)

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Table 1 — *continued from previous page*

#	Benchmark	Variant	Sub-RQ	What it differentiates
10	Split-CIFAR ₁₀	HN, grad only	2C2	Is unit- ℓ_2 gradient normalization alone sufficient to prevent inversion collapse, or is Fisher abs-max scaling also required?
11	Split-CIFAR ₁₀₀	Standard HN	1b	Does the Sub-RQ ₁ finding (Adam dominates in the hypernetwork setting) hold as task count doubles to 10 and per-task classification difficulty increases to 10 classes?
12	Split-MNIST (SH)	HN $d_h=4$	App.	<i>Prelim. sweep lower bound.</i> Does $d_h=4$ cause complete training failure across all seeds? Justifies the choice of $d_h=8$ in Config 7.
13	Split-MNIST (SH)	HN $d_h=16$	App.	<i>Prelim. sweep upper bound.</i> Does $d_h=16$ produce a ceiling (≥ 0.93) with no method separation? Justifies the choice of $d_h=8$ in Config 7.
14	Permuted-MNIST	EMA Ac-cum.	5	Stress Test. Does Exponential Moving Average (EMA) Fisher accumulation improve retention over MAX accumulation in a long 20-task sequence?
15	Permuted-MNIST	MAX Ac-cum.	5	Stress Test. Paired baseline for Config 14 using standard MAX Fisher construction on a 20-task sequence.
16	Split-MNIST (SH)	HN Sweep	β Ablat.	<i>Sweep A.</i> Determines stable β penalty weights for the suf-focated hypernetwork regime ($c=1000$).
17	Split-CIFAR ₁₀	HN Sweep	β/c Ablat.	<i>Sweep B.</i> Identifies the optimal trade-off between backward transfer (BWT) and accuracy by sweeping β and chunk size (c).
18	Split-CIFAR ₁₀₀	HN Sweep	β Ablat.	<i>Sweep C.</i> Evaluates stability under high gradient explosion risk for massive target networks ($c=6000$).

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Table 1 — *continued from previous page*

#	Benchmark	Variant	Sub-RQ	What it differentiates
19	Split-MNIST (SH)	SGD sweep	lr	Diag. Diagnostic. At what learning rate does SGD fail to produce a usable task-1 initialisation on Split-MNIST within the 5-epoch budget? Establishes that the convergence threshold lies between 5×10^{-5} and 10^{-4} . Critically, the paper’s own FOPNG learning rate for Split-MNIST is $\eta=10^{-5}$, which lies a full decade below this threshold (SGD at 10^{-5} yields acc ≈ 0.21 , barely above random). This directly falsifies the paper’s stated first-task protocol “SGD at the same learning rate as the optimizer” for FOPNG on Split-MNIST.
20	Split-CIFAR10 (MH)	SGD sweep	lr	Diag. Diagnostic. Identifies the SGD convergence threshold for Split-CIFAR10 MH within 5 epochs. Result: the paper’s stated $\eta=10^{-3}$ yields only $\approx 85\%$ task-1 accuracy with loss $\approx 0.65 > \ln 2$, far below the $\approx 95\%$ shown in Figure 1 of Garg et al. (n.d.). Threshold is $\geq 10^{-2}$, directly refuting the paper’s claimed first-task protocol for CIFAR-10 experiments.

Table 2: Experiment inventory — execution details. **Method sets:** ALL = {iFOPNG, FOPNG, OGD, ONG, FNG, EWC, Adam, SGD}; PROJ = {iFOPNG, FOPNG, OGD, ONG, FNG, EWC}; FISHER = {iFOPNG, FOPNG, OGD, FNG}; MINI = {iFOPNG, Adam}. “Table 1” = Garg et al. (2026) Table 1 exactly (per-method lr and λ). “HN” = uniform lr = 10^{-3} , $\lambda = 10^{-3}$, Fisher = 1024, batch = 64 (see Section 4.5.1). *EWC in Config 6 uses Fisher = 5000 (Table 1 “full”); all other methods use 1024.

#	Benchmark	First task	Methods	Seeds	Notable parameters	Status
1	Permuted-MNIST	SGD method lr	@	ALL	3 Table 1 HPs. Fisher = full (60K). 5 tasks $\times$ 5 ep.	✓
2	Split-MNIST (MH)	SGD method lr	@	ALL	3 Table 1 HPs. Fisher = full (12K). 5 tasks $\times$ 5 ep. Multi-head 5×2 .	✓
3	Split-MNIST (SH)	SGD method lr	@	ALL	3 Table 1 HPs. Fisher = full (12K). 5 tasks $\times$ 5 ep. Single 10-class head.	✓

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Table 2 — *continued from previous page*

#	Benchmark	First task	Methods	Seeds	Notable parameters	Status
4	Split-CIFAR ₁₀ (MH)	SGD @ 10^{-3}	ALL	3	Table 1 HPs. Fisher = 1024. 5 tasks $\times$ 5 ep. Multi-head 5×2 .	✓ (verify)
5	Split-CIFAR ₁₀ HN	Adam @ 10^{-3}	FISHER	3	HN (64/32/32/256, 185 chunks). $\beta = 0.01$. 5 tasks $\times$ 50 ep. Full normalization.	✓
6	Split-CIFAR ₁₀₀ (MH)	SGD @ 10^{-2}	ALL	3	Table 1 HPs. Fisher = 1024 (EWC: 5000*). max_dirs = 800. 10 tasks $\times$ 10 ep.	✓
7	Split-MNIST SH HN	AdamW 10^{-3}	@ ALL	5	Suffocated HN ($d_h=8$, emb = 32/32, chunk = 64, 1,406 chunks). 5 tasks $\times$ 15 ep. Full normalization.	✓
8	Split-CIFAR ₁₀ HN	AdamW 10^{-3}	@ ALL	3	Standard HN (64/32/32/256, 185 chunks). $\beta = 0.01$. 5 tasks $\times$ 50 ep. Full normalization.	✓
9	Split-CIFAR ₁₀ HN	AdamW 10^{-3}	@ iFOPNG	3	Config 8 settings. No -normalize flag. Log cond($G^\top \hat{F}G$) per step.	It didnt crash for ifopng. The efficiency went down.
10	Split-CIFAR ₁₀ HN	AdamW 10^{-3}	@ iFOPNG	3	Config 8 settings. -normalize_gradients_only (unit ℓ_2 cols; raw Fisher).	TODO DEMANDS SMALL SPECIFIC CHANGES. NO FLAG FOR IT.

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Table 2 — *continued from previous page*

#	Benchmark	First task	Methods	Seeds	Notable parameters	Status
11	Split-CIFAR ₁₀₀ HN	AdamW 10^{-3}	@ ALL	3	Standard HN (64/32/32/256). Same architecture as Config 8. 10 tasks $\times$ 50 ep. Full normalization.	✓ (SGD OGD off. incorporate OGD 1e-4 lr)
12	Split-MNIST SH HN	AdamW 10^{-3}	@ MINI	2	$d_h=4$, emb = 32/32, chunk = 64. 5 tasks $\times$ 15 ep. Result: Not a total collapse but a degrading performance nonetheless of projection methods.	✓
13	Split-MNIST SH HN	AdamW 10^{-3}	@ MINI	2	$d_h=16$, emb = 32/32, chunk = 64. 5 tasks $\times$ 15 ep. Result: Both projections and Adam do similarly.	✓
14	Permuted-MNIST	SGD @ 10^{-3}	iFOPNG EMA	5	20 tasks $\times$ 5 ep. Fisher = 60000. $\alpha = 0.5$. Config 414.	✓
15	Permuted-MNIST	SGD @ 10^{-3}	iFOPNG	5	20 tasks $\times$ 5 ep. Fisher = 60000. $\alpha = 0.5$. Config 415.	✓
16	Split-MNIST SH HN	AdamW 10^{-3}	@ Adam	1	Exp 701. $c = 1000$, $d_h = 8$, emb = 4/10. Sweeps $\beta \in \{0.01, 0.05, 0.1\}$.	✓
17	Split-CIFAR ₁₀ HN	AdamW 10^{-3}	@ Adam	1	Exp 702. $d_h = 32$, emb = 16/16. Sweeps β and $c \in \{64, 500, 1000, 2000, 5000, 6000\}$.	✓
18	Split-CIFAR ₁₀₀ HN	AdamW 10^{-3}	@ Adam	1	Exp 703. Fast check (5 tasks $\times$ 15 ep). $c = 6000$, $d_h = 128$. Sweeps β .	✓

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Table 2 — *continued from previous page*

#	Benchmark	First task	Methods	Seeds	Notable parameters	Status
19	Split-MNIST (SH)	SGD @ sweep lr	SGD	5	Diagnostic. Exp 419. 1 task $\times$ 5 ep. Sweeps $\eta \in \{10^{-5}, 5 \times 10^{-5}, 10^{-4}, 2 \times 10^{-4}, 5 \times 10^{-4}, 10^{-3}, 5 \times 10^{-3}, 10^{-2}\}$. Fisher = 12000. Batch = 10. Result: threshold between 5×10^{-5} and 10^{-4} . Paper's FOPNG lr = $10^{-5} \Rightarrow$ SGD acc ≈ 0.21 (barely above random); protocol is self-contradictory.	✓
20	Split-CIFAR ₁₀ (MH)	SGD @ sweep lr	SGD	5	Diagnostic. Exp 420. 1 task $\times$ 5 ep. Sweeps $\eta \in \{10^{-3}, 5 \times 10^{-3}, 10^{-2}, 5 \times 10^{-2}\}$. Fisher = 1024. Batch = 10. Result: paper's $\eta = 10^{-3}$ yields acc ≈ 0.85 , loss ≈ 0.65 (above random ln 2). True convergence only at $\eta \geq 10^{-2}$; refutes paper's first-task protocol for CIFAR-10.	✓

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